

Halloween Crossword Solutions

Kevin Chen, Sohum Uttamchandani

¹ 1	² 3	³ 5
⁴ 0	1	3
⁵ 2	8	8

ONE ACROSS

First, choose which 2 monsters receive their own correct cloaks:

$$\binom{6}{2} = 15.$$

The remaining 4 monsters must all receive incorrect cloaks. The number of derangements of 4 objects is

$$!4 = 9.$$

Therefore, the total number of redistributions is

$$15 \cdot 9 = \boxed{135}.$$

ONE DOWN

Let the numbers of candy eyeballs received by the vampire, werewolf, and mummy be x , y , and z . Since each monster receives at least one,

$$x + y + z = 16, \quad x, y, z \geq 1.$$

Without the vampire's restriction, the number of positive integer solutions is

$$\binom{15}{2} = 105.$$

Now count the invalid distributions where the vampire receives at least 13 candy eyeballs. Let

$$x' = x - 12.$$

Then

$$x' + y + z = 4$$

with all three variables positive, giving

$$\binom{3}{2} = 3$$

invalid distributions.

Thus, the number of valid distributions is

$$105 - 3 = \boxed{102}.$$

TWO DOWN

By Vieta's formulas,

$$a + b + c = 30$$

and

$$ab + bc + ca = 291.$$

Using the identity

$$a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca),$$

we obtain

$$a^2 + b^2 + c^2 = 30^2 - 2(291) = 900 - 582 = \boxed{318}.$$

THREE DOWN

Let the number of glowing skulls be n . The conditions give

$$n \equiv 6 \pmod{7}, \quad n \equiv 7 \pmod{9}, \quad n \equiv 10 \pmod{11}.$$

The first and third congruences can be written as

$$n \equiv -1 \pmod{7}$$

and

$$n \equiv -1 \pmod{11}.$$

Since 7 and 11 are relatively prime,

$$n \equiv -1 \pmod{77}.$$

Thus,

$$n = 77k - 1.$$

Using the condition modulo 9,

$$77k - 1 \equiv 7 \pmod{9}.$$

Since $77 \equiv 5 \pmod{9}$,

$$5k \equiv 8 \pmod{9}.$$

Multiplying by the inverse of 5 modulo 9, which is 2, gives

$$k \equiv 7 \pmod{9}.$$

Since $n < 693$, we take $k = 7$. Therefore,

$$n = 77(7) - 1 = 539 - 1 = \boxed{538}.$$

FOUR ACROSS

We want to evaluate

$$17^{103} \pmod{100}.$$

Since

$$17^{20} \equiv 1 \pmod{100},$$

we reduce the exponent modulo 20:

$$103 \equiv 3 \pmod{20}.$$

Therefore,

$$17^{103} \equiv 17^3 \pmod{100}.$$

Now,

$$17^3 = 4913 \equiv 13 \pmod{100}.$$

Written as a three-digit crossword entry, the answer is

$$\boxed{013}.$$

FIVE ACROSS

We use the identity

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - (xy + yz + zx)).$$

First,

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx).$$

Substituting the given values,

$$x^2 + y^2 + z^2 = 12^2 - 2(44) = 144 - 88 = 56.$$

Therefore,

$$x^3 + y^3 + z^3 - 3(48) = 12(56 - 44).$$

So,

$$x^3 + y^3 + z^3 - 144 = 144,$$

and hence

$$x^3 + y^3 + z^3 = \boxed{288}.$$